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1. Which of the following scheduling algorithms gives minimum average waiting time?
(a) shortest job first (b) Round robin
(c) priority (d) First come first serve

Computer – CPU Scheduling – 2021

2. A CPU –bound program might have
(a) a few long CPU bursts
(b) a few short CPU bursts
(c) many short CPU bursts
(d) None of the above

Computer – Operating System (CPU Bound) – 2021

3. The amount of time required to read a block of data from a disk into memory is composed of seek time, rotational latency and transfer time. Rotational latency refers to
(a) the time its taken for the platter to make a full rotation
(b) the time its taken for the read-write head to more into position over the appropriate track
(c) the time it taken for the platter to rotate the correct sector under the head
(d) to reduce number of bits in the field of instruction.

COMPUTER – Disk Scheduling – 2023

4. How many child processes will be created by following fork ( ) system call?
fork();
fork();
fork();
fork();
(a) 4 (b) 16 (c) 15 (d) 3

Operating System – Process Creation (fork()) - 2024

5. The current allocation and maximum requirement of different types of resources for four processes are given below:

Process	Max			Allocation			Available		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P_1	8	6	4	1	2	1	4	4	5
P_2	4	3	3	3	1	1			
P_3	10	1	3	4	1	3			
P_4	3	3	3	3	2	2			

Consider the following four statements
(A) P2-- >P4 -- >P1-- >P3 is a safe sequence
(B) P4-- >P2 -- >P1-- >P3 is a safe sequence.
(C) P4-- >P2 -- >P3-- >P1 is a safe sequence
(D) P1-- >P4 -- >P2-- >P3 is a safe sequence.
Identify correct statements from the given options.
(a) A, B and D only (b) A, B and C only
(c) B, C and D only (d) A, B, C & D

Operating System – Banker's Algorithm / Safe Sequence (Deadlock Avoidance) - 2024

6. An operating system contains 4 user processes each requiring 5 units of resource R. The minimum number of required units of R such that no deadlock will ever occur is
(a) 20 (b) 4 (c) 17 (d) 15

Operating System – Deadlock (Minimum Resource Requirement) - 2024

7. Match List-I with List-II

List-I		List-II	
A.	Critical Region	I.	Circular wait
B.	Working Set	II.	Condition variable
C	Deadlock	III.	Principle of locality
D.	Wait/signal	IV.	Mutual exclusion

Choose the correct answer from the options given below
(a) A-IV, B-III, C-I, D-II (b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I (d) A-III, B-IV, C-I, D-II

Operating System – Deadlock & Synchronization - 2024

8. In a computer if the page fault service time is 10 ms and average memory access time is 30 ns. If one page fault is generated for every 106 memory accesses. What is the effective access time for the memory?
(a) 21 ns approximate (b) 25 ns approximate
(c) 30 ns approximate (d) 40 ns approximate

Operating System – Paging (Effective Memory Access Time) - 2024

9. How does the number of page frames affect the number of page faults for a given memory access pattern in FIFO page replacement algorithm?
(a) Increasing the number of page frames always decreases the number of page faults.
(b) Increasing the number of page frames may increase or decrease the number of page faults depending on the memory access pattern.
(c) Increasing the number of page frames always increases the number of page faults.
(d) Increasing the number of page frames has no effect on the number of page faults.

Operating System – Page Replacement (FIFO & Belady's Anomaly)-2024

10. On a system using simple segmentation, following is the segement table

Segment	Limit	Base
0	500	1000
1	200	2000
2	300	2500
3	100	1700

What is the physical address for the logical address 2, 212?
(a) 2712 (b) 512
(c) 2212 (d) 2800

Operating System – Segmentation (Logical → Physical Address)

11. Decreasing the RAM of a computer typically leads to which of the following outcomes?
(a) Virtual memory increases
(b) Page faults increases
(c) Page faults decreases
(d) Segmentation faults occur
Operating System – Virtual Memory (Effect of RAM on Page Faults)-2024

12. Consider a system with 1K pages and 512 frames and each page is of size 2KB. How many bits are required to represent the virtual address space memory.
(a) 20 bits (b) 21 bits
(c) 11 bits (d) 16 bits
Operating System – Virtual Address Space*2024

13. Arrange the following layers of MS DOS operating system starting from inner most to outer most.
(a) ROM BIOS Device drivers
(b) Resident System Program
(c) MS DOS Device Drivers
(d) Application Program
Choose the **correct** answer from the options given below:
(CUET-PG 2025) CPU Scheduling
(a) (A), (C), (B), (D)
(b) (D), (B), (C), (A)
(c) (B), (A), (D), (C)
(d) (C), (B), (D), (A)

14. Which of the following scheduler/schedulers is/are also called CPU scheduler?
(CUET-PG 2025) CPU Scheduling
(A) Short Term Scheduler
(B) Long Term Scheduler
(C) Medium Term Scheduler
(D) Asymmetric Scheduler
Choose the **correct** answer from the options given below:
(a) (A), (B) and (D) only
(b) (A), (B) and (C) only
(c) (A), (B), (C) and (D)
(d) (A) only

15. Consider a page refrence string as : 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, Assume that there are 3-page frames available. calculate the total number of pages faults for the above reference string if **LRU** policy is used for page replacement.
(CUET-PG 2025) Page Replacement
(a) 10 (b) 8 (c) 9 (d)11

16. A situation where two or more processes are blocked, waiting for resources held by each other is called:
(CUET-PG 2025) Deadlock
(a) Pooling (b) Deadlock
(c) Thrashing (d) Paging

17. Which of the following is a way to recover from the deadlock which has already occurred?
(CUET-PG 2025) Deadlock
(a) Process termination
(b) Non preemption of resources
(c) Banker's algorithm
(d) Circular wait

18. Complete the following statement by choosing the correct option.
For a deadlock to occur, the four conditions namely Mutual Exclusion, Hold and wait, No preemption, circular wait_______
(CUET-PG 2025) Deadlock
(a) May or may not hold
(b) The circular wait does not implies hold and wait condition
(c) Are completely independent
(d) Must hold simultaneously

19. Match **List-I** with **List-II**

List-I (CPU Scheduling Algorithm)		List-II (Feature)	
(A)	FCFS	(I).	FCFS + preemption
(B)	Round Robin	(II).	Allows the processes to move between queses
(C)	Multi level queue scheduling	(III).	Often long average waiting time
(D)	Multi level Feedback Queue	(IV).	Permanent assignment of processes to one specific queue

Choose the **correct** answer from the options gven below:
(CUET-PG 2025) CPU Scheduling
(a) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
(b) (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
(c) (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
(d) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

20. Which disk scheduling algorithm looks for the track closest to the current head position?
(CUET-PG 2025) Disk Scheduling
(a)LOOK (b)SSTF
(c)FCFS (d)SCAN

21. Match **List-I** with **List-II**

	List-I		List-II
(A)	Seek Time	(I).	Total number of bytes transferred, divided by the total time between the first request for service and the completion of the last transfer
(B)	Access Time	(II).	Time taken for the disk to rotate the desired sector to the disk head
(C)	Rotational Latency	(III)	Seek Time + Rotational Latency
(D)	Disk Bandwidth	(IV)	Time taken for the disk arm to move the heads to the cylinder containing the desired sector

(CUET-PG 2025) Disk Scheduling

Choose the **correct** answer from the options gven below:

(a) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
(b) (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
(c) (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
(d) (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

22. Which CPU scheduling algorithm prefers the process with the shortest burst time?

(CUET-PG 2025) Scheduling

(a) FCFS (b) SJF
(c) Round Robin (d) Priority Scheduling

23. Consider a typical process P, in the critical section. Arrange the following statements of code to make a valid general structure.

(CUET-PG 2025) Process Section

(A) Critical section (B) Remainder section
(C) Entry section (D) Exit section

Choose the correct answer from the options given below:

(a) (C) (b) (A), (D) (b) (c) (A), (b) (D)
(c) (C), (A), (D), (B) (d) (C), (b) (D), (A)

24. Consider the following type of memories.

(A) Hard Disk Drive (HDD)
(B) Cache Memory
(C) Random Access Memory
(D) Register

Arrange the above memories according to their access speed (from fastest to slowest)

(CUET-PG 2025) Memory Hierarchy

(a) (D), (C), (B), (A) (b) (A), (C), (B), (D)
(c) (B), (D), (C), (A) (d) (D), (B), (C), (A)

25. 'The Dining Philosopher' problem can be solved by:

(CUET-PG 2025) Operating System Syncronization

(a)Use of semaphores (b)Use of overlays
(c)Mutual exclusion (d)Bounded waiting

26. Details a paging system for memory management are as follows:

Logical address space: 32 KB
Page Size: 4 B
Physical Memory size: 64 KB

The number of pages in the logical address space and number of page frames in physical memory, respectively are:

(CUET-PG 2025) Paging

(a) 4, 16 (b) 4, 12
(c) 8, 16 (d) 3, 16

27. External fragmentation occurs

(CUET-PG 2025) Meemory Management

(a) When enough total memory space exists to satisfy a request, but it is not contiguous, storage is fragmented into a large number of small holes.
(b) When enough total memory space exists to satisfy a request, which is contiguous, storage is fragmented into a large number of small holes.
(c) When memory is empty.
(d) Always.

ANSWER KEY

1.	2.	3.	4.	5.	6.	7.	8.	9.	10.
a	a	c	c	b	c	a	d	b	a
11.	12.	13.	14.	15.	16.	17.	18.	19.	20.
b	b	a	d	c	b	a	d	b	b
21.	22.	23.	24.	25.	26.	27.			
d	b	c	d	a	c	a			